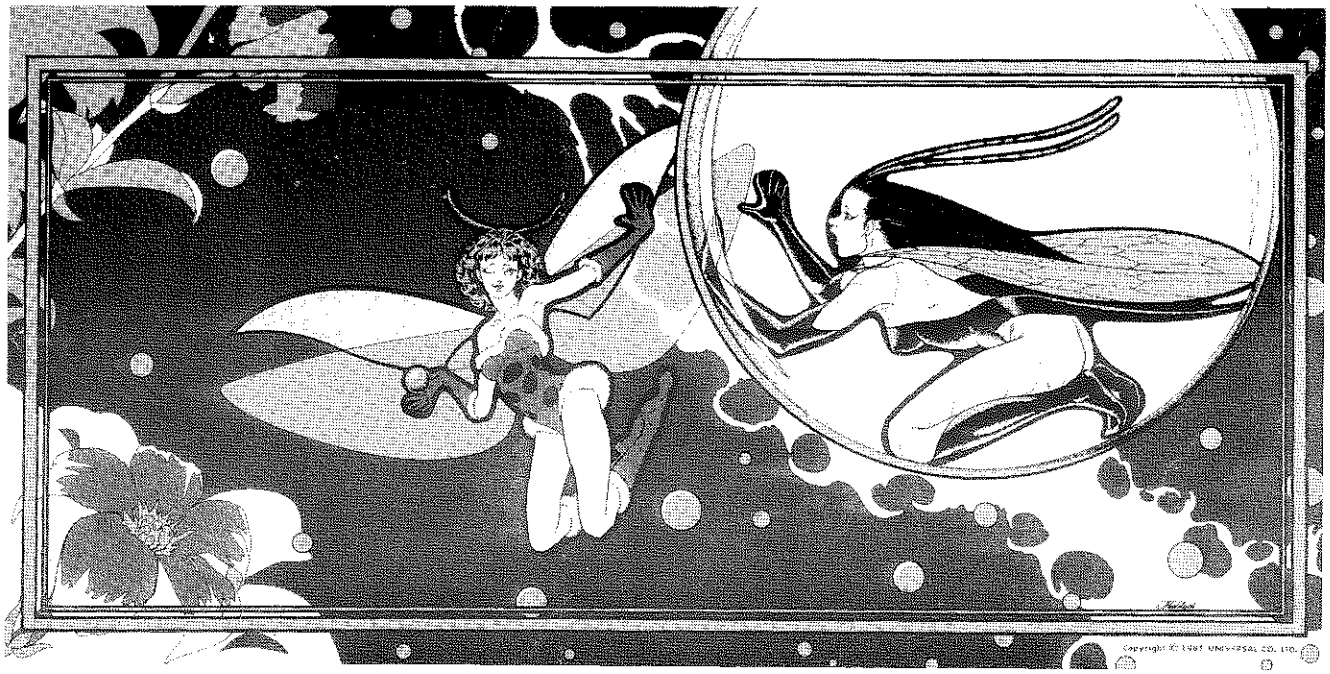
Lo 061348 Bug

Operation, Maintenance

and Service Manual



UNIVERSAL

"WARNING: This equipoent generates, uses, and can radiate radio frequency energy and if not installed and used in accordance with the instructions manual, may cause interference to radio communications. As temporarily permitted by regulation it has not been tested for compliance with the limits for Class A computing devices pursuant to Subpart J of Part 15 of FCC Rules, which are designed to provide reasonable protection against such interference. Operation of this equipment in a residential area is likely to cause interference in which case the user at his own expense will be required to take whatever measures may be required to correct the interference."

Thank you very much for your purchase of the LADY BUG. Carrying a CPU, the LADY BUG is a game machine of the newest type with many features and functions.

This manual describes how to make the most of such features and functions, so please read it carefully in order to make the best use of the machine.

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— WHEN ORDERING PARTS —

Since each component part is indicated by block, definitely specify both the corresponding Fig. No. and part No. within the Fig. when placing an order for it.

I. FEATURES OF THIS MACHINE

1. This video game packs in a variety of excellent features captivating players.

2. The lever control system is simple to operate.

3. By setting or combining dip switches one can select anyone of a variety of game features.

4. Additionally, by means of dip switches, one determines which of the four (4) levels of difficulty desired.

5. The highest scorer for the day is always displayed on the screen, and the 9 highest scorers can enter their names on the screen as well (dip switch combinations are also used to select letters when resistering the player's name).

6. Ultra-contemporary cabinet and fascinating acoustics.

II. HOW TO HANDLE AND MAINTAIN THIS MACHINE

1. Since the UNIVERSAL's LADY BUG employs a color TV receiving set, be careful not to shake it during transit and when carrying it about.

2. Install it at a location which is not exposed to direct sunlight. In order to prevent the inside temperature rising, avoid as much as possible a location near a heater, etc.

3. Since the grounding terminal is visible, be sure to connect it to a grounding conductor.

4. Insert the power cord into the outlet and turn on the switch.

5. Even if the solid-state module seems to be out of order, do not check the circuit by means of a circuit tester, etc., since the internal voltage of the tester, etc. may sometimes break down the IC.

6. Make sure the machine is well ventilated. If the temperature of the IC and transistor is lower than 60°C, the function normally and may be considered reliable. If it exceeds 60°C, their performance cannot be guaranteed.

7. Make sure that the connector, etc. are not disconnected.

8. Whenever connecting the power cord of the solid-state module to, or disconnecting it from, the outlet, be sure to turn the power off.

9. Although the products of UNIVERSAL are manufactured with the utmost care, they may develop malfunctions when used for long periods. So, be sure to check this machine daily.

III. HOW TO PLAY

1. By working the lever, move the "Lady Bug" and wipe out all dots (□) on the course.
2. When the "Lady Bug" hits the scull mark or is eaten by a predatory enemy insect, it disappears.
3. Predatory enemy insects will chase the "Lady Bug" around the screen. Use the turnstile to allow the "Lady Bug" to dodge the enemy. (The enemy insects cannot turn the turnstile.)
4. If you skillfully operate the turnstile you can change course easily, effectively dodging the enemy assault.
5. When the "Lady Bug" eats "vegetable dots" appearing at the center of the screen, a high score is awarded.
6. When players lose all the "Lady Bugs", the game is over.
7. 1st – 9th highest scores for the day can enter their names on the screen (the number of letters can be changed by dip switches).

SCORE

- Point dot (□) ..... 10 PTS
- Feature dot ( ) ..... 100 PTS / 300 PTS / 800 PTS
- Vegetable dot ( ) ..... 1000 ~ ?

FEATURES

SPECIAL When all S. P. E. C. I. A. L. letters have been eaten up and if the feature dots remain red, players are awarded another game.

EXTRA When all E. X. T. R. A. letters have been eaten up and if the feature dots remain yellow, players are awarded another "Lady Bug".

BONUS When the ♡ mark has been eaten up and if the feature dots remain blue, the dot score multiplies by 2X, 3X, 5X.

IV. VARIOUS OPTIONAL SETTINGS

A. SERVICE SWITCHES AND DIP SWITCHES

- **Dip Switch A**

1. Number of Lady Bugs Given (SW1)

No. of Lady Bugs	SW1
3	OFF
5	ON

2. Free Play (SW2)

Free Play	SW2
Need Coin	OFF
Free Play	ON

3. Type (SW3)

Type	SW3
Table Type	OFF
Upright Type	ON

4. Screen Stoppage (SW4)

Screen	SW4
Normal	OFF
Screen Stoppage	ON

5. Automatic Renewal of Screen (SW5)

Automatic Screen	SW5
Normal	OFF
Auto Renewal of Screen	ON

6. High Score (SW6)

Score	SW6
Full Letters	OFF
0 – 3 Letters	ON

7. Class (SW7, 8)

Class	SW7	SW8
First Class	OFF	OFF
Second Class	OFF	ON
Third Class	ON	OFF
Top Class	ON	ON

- **Positions of Volume Controller, Dip and Service Switches**

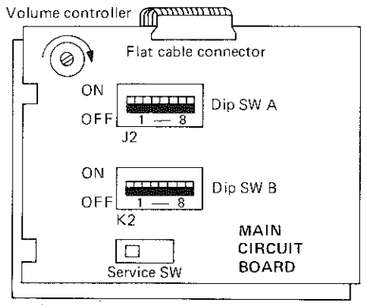


Fig. 1 Positions of Volume Controller, Dip and Service Switches

Service Switch
Pressing the service switch, the left-hand chute is set and the number of credit/s increases.

- **Dip Switch B (Coin & Credit)**

Coin	Credit	Lift-hand Chute				Right-hand Chute			
		SW1	SW2	SW3	SW4	SW5	SW6	SW7	SW8
1	1	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
1	2	OFF	OFF	OFF	ON	OFF	OFF	OFF	ON
1	3	OFF	OFF	ON	OFF	OFF	OFF	ON	OFF
1	4	OFF	OFF	ON	ON	OFF	OFF	ON	ON
1	5	OFF	ON	OFF	OFF	OFF	ON	OFF	OFF
2	1	OFF	ON	OFF	ON	OFF	ON	OFF	ON
2	3	OFF	ON	ON	OFF	OFF	ON	ON	OFF
3	1	OFF	ON	ON	ON	OFF	ON	ON	ON
3	2	ON	OFF	OFF	OFF	ON	OFF	OFF	OFF
4	1	ON	OFF	OFF	ON	ON	OFF	OFF	ON
1	1	ON	OFF	ON	OFF	ON	OFF	ON	OFF
1	1	ON	OFF	ON	ON	ON	OFF	ON	ON
1	1	ON	ON	OFF	OFF	ON	ON	OFF	OFF
1	1	ON	ON	OFF	ON	ON	ON	OFF	ON
1	1	ON	ON	ON	OFF	ON	ON	ON	OFF
1	1	ON	ON	ON	ON	ON	ON	ON	ON

Standard and Custom Price Settings

The game price set by a combination of dip SWs is displayed on the monitor when coin credit is 0.

a) When the settings for right and left chutes are the same, "1 coin – 1 credit", "2 coins – 3 credits", etc. are displayed.

b) When the settings for right and left chutes are provided differently, their respective contents can be displayed.

V. TV MONITOR

A. ADJUSTMENT OF TV PICTURE SCREEN

[UPRIGHT TYPE]

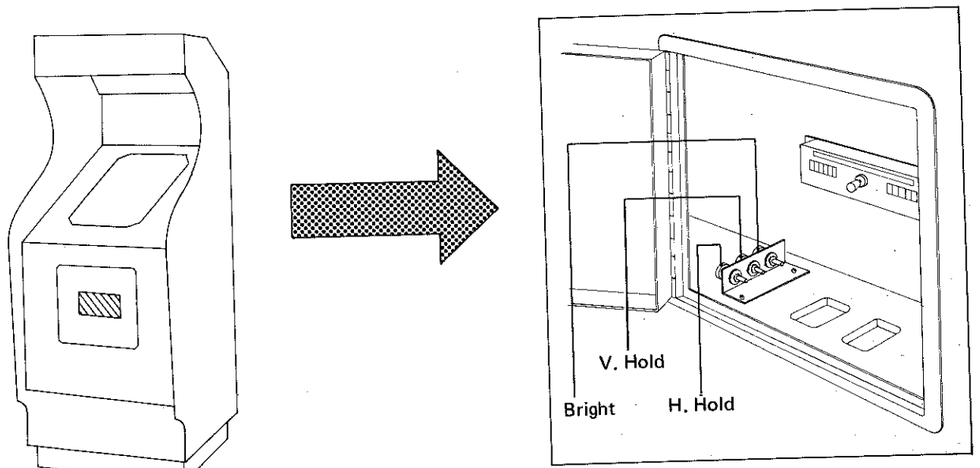


Fig. 2 Positions of TV Monitor Controller (Upright Type)

[TABLE TYPE]

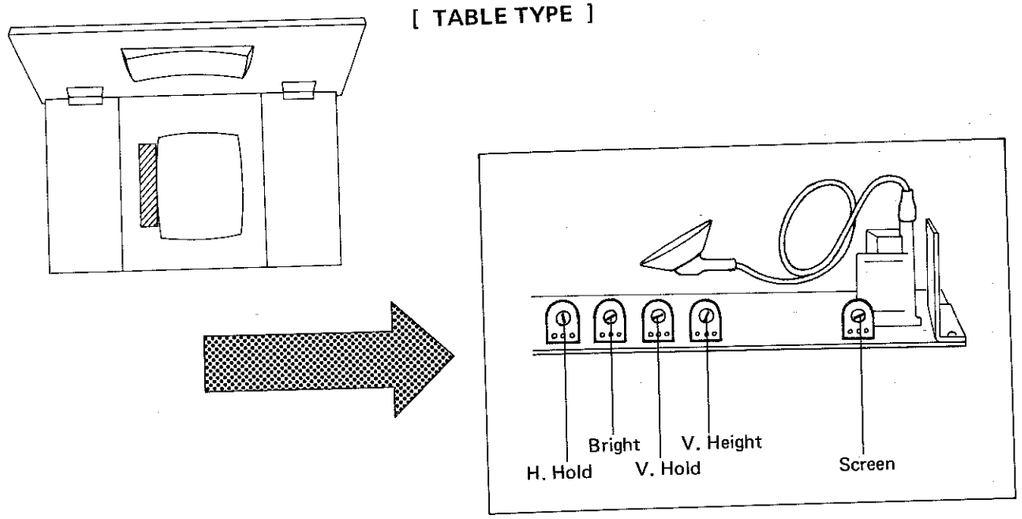
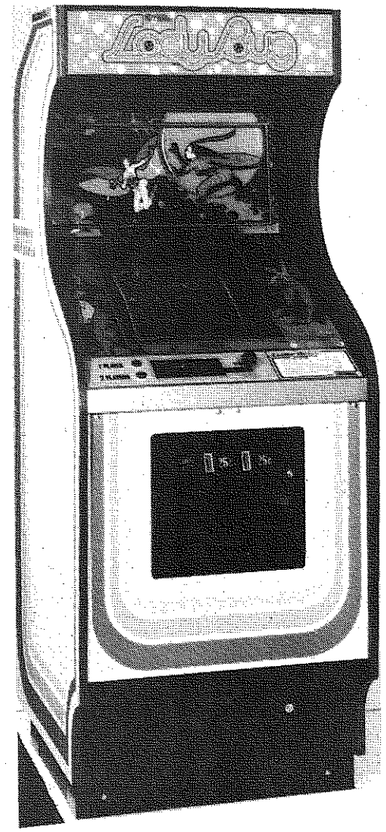


Fig. 3 Positions of TV Monitor Controller (Table Type)

UPRIGHT TYPE



SPECIFICATIONS
▽95-1653
800(D) x 640(W) x 1750(H) mm
AC100V/115V/230V 50/60Hz
180W (20")

This documentation provides standard information.
Universal reserves the right to change without notice.

VI. UPRIGHT TYPE PARTS CATALOG

A. COMPONENT PARTS RELATED TO CABINET (OUTSIDE)

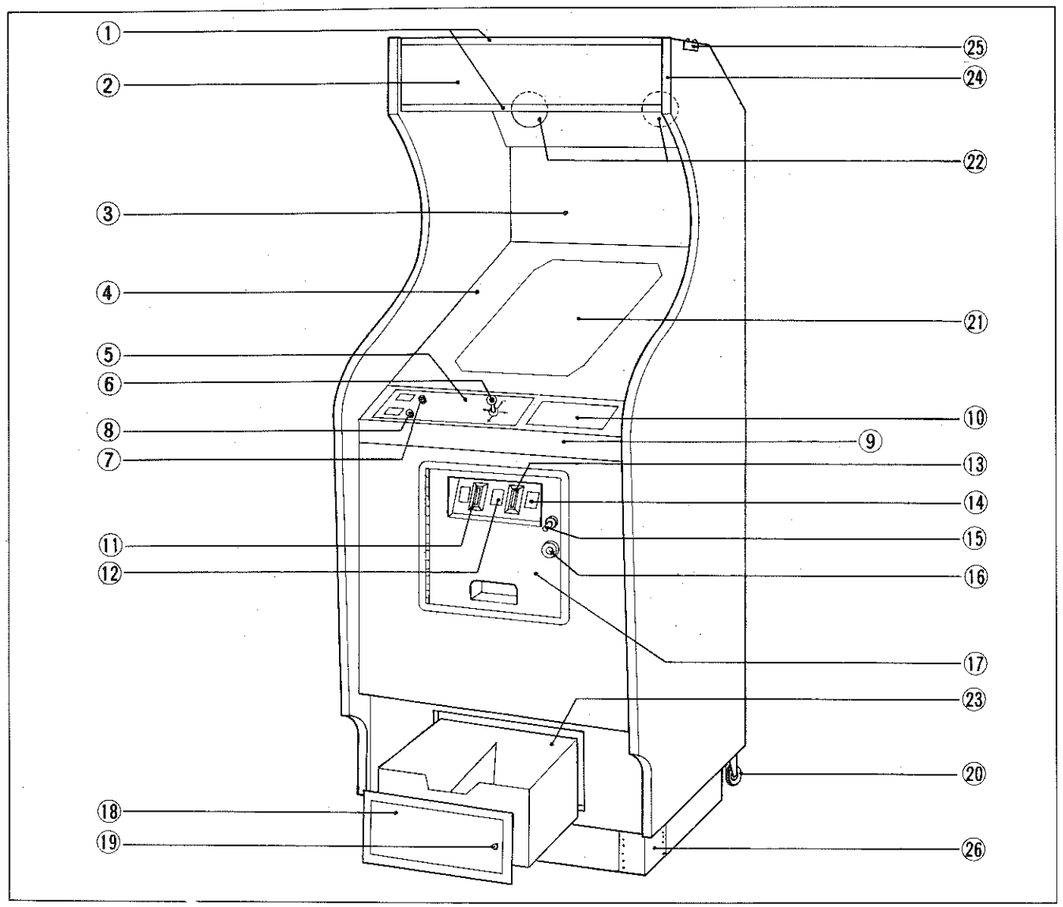


Fig. 4 Component Parts Related to Cabinet (Outside)

OUTSIDE CABINET PARTS LIST

Ref. No.	Name	Ref. No.	Name
1	Title panel fixture (Twin)	14	Coin indication panel (2)
2	Title panel (153 x 596mm)	15	Cancel button
3	Illustrated glass (B) (295 x 596mm)	16	Door key
4	Illustrated glass (A) (566 x 596mm)	17	Main door
5	Operating indication panel (110 x 400mm)	18	Cash box door
6	Control lever (general name)	19	Cash box door key
7	Push button (for 1 player)	20	Caster
8	Push button (for 2 players)	21	CRT (20" color)
9	Operating base plate	22	Speaker (x2)
10	Sticker for game rules (110 x 180mm)	23	Cash box
11	Coin slot (1)	24	T-Moulding
12	Coin indication panel (1)	25	Hook metal
13	Coin slot (2)	26	L-shaped reinforcing metal fittings

B. DRAWING OF CIRCUIT BOARD MOUNTING POSITIONS

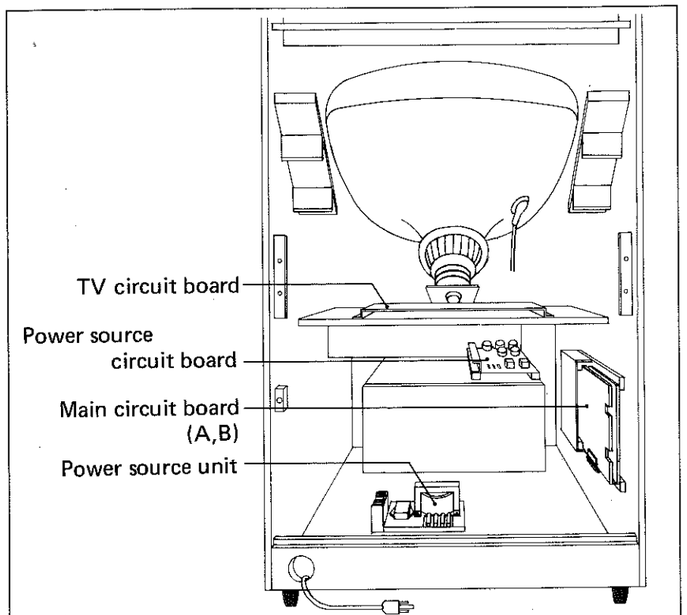


Fig. 5 Circuit Board Mounting Positions

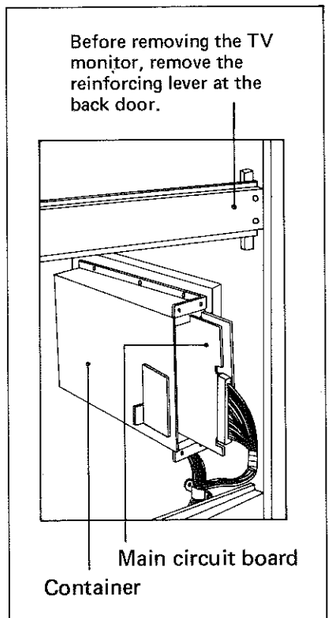


Fig. 6 Side View of Circuit Board Mounting Positions

C. FUSES

Fuses in the power source unit

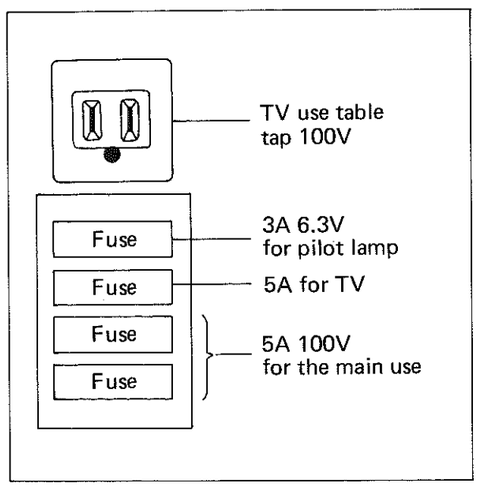


Fig. 7 Fuses in the Power Source Unit

Fuses in the Power Source Circuit Board

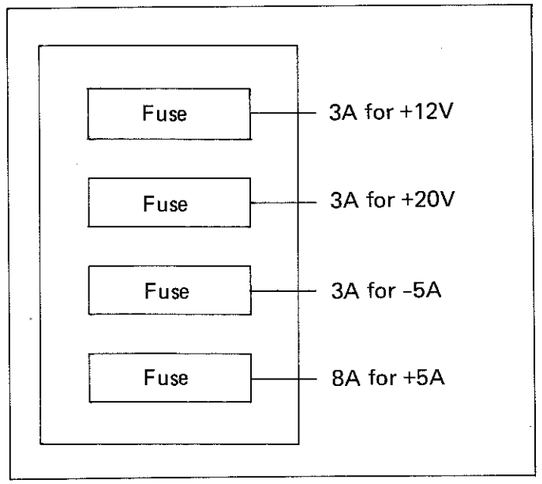


Fig. 8 Fuses in the Power Source Circuit Board

D. COMPONENT PARTS RELATED TO CABINET (INSIDE)

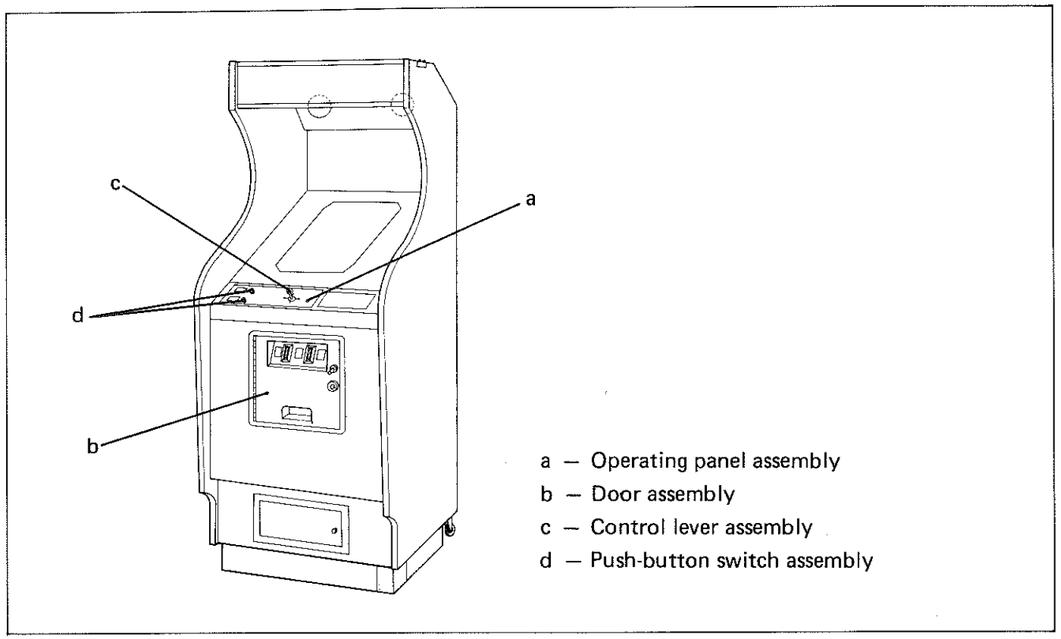


Fig. 9 Main Assemblies Related to Cabinet (Inside)

a) Operating panel assembly

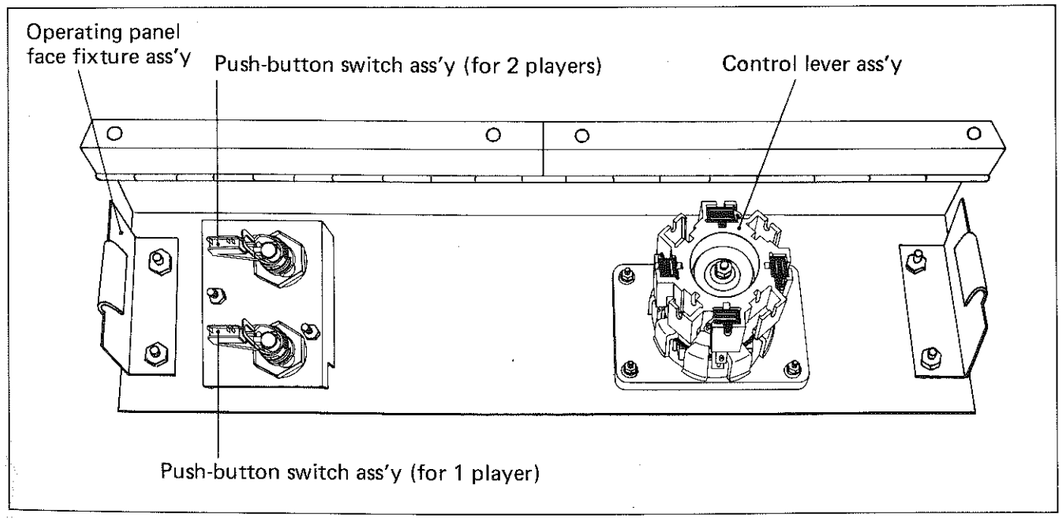


Fig. 10 Operating Panel Assembly

b) Door assembly and parts list

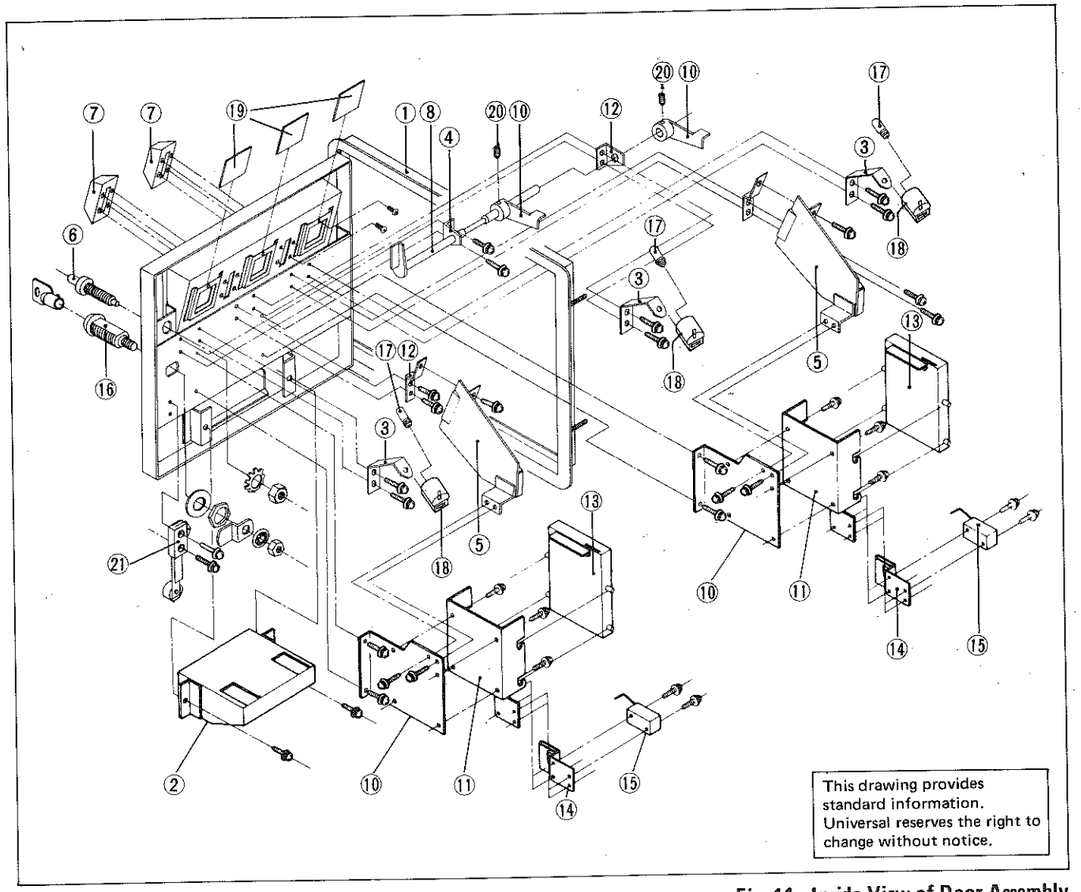


Fig. 11 Inside View of Door Assembly

DOOR ASSEMBLY PARTS LIST

Ref. No.	Name	Ref. No.	Name
1	Main door	12	Coin slot chute holder (x 2)
2	Returning saucer	13	Rejector (x 2)
3	Lamp bracket (x 3)	14	Sensor slot (x 2)
4	Rearing (x 2)	15	Micro switch
5	Coin slot chute (x 2)	16	Key sets
6	Returning button	17	Miniature lamp (x 3)
7	Coin slot (x 2)	18	Miniature lamp socket (x 3)
8	Transmission shaft	19	Coin indication panel (25ϕ x 2, coin slot)
9	Rotary bracket (x 2)	20	Hexagon socket head screw (x 2)
10	Rejector bracket (x 2)	21	Tilt switch
11	Rejector case (x 2)		

c) Control lever assembly and parts list

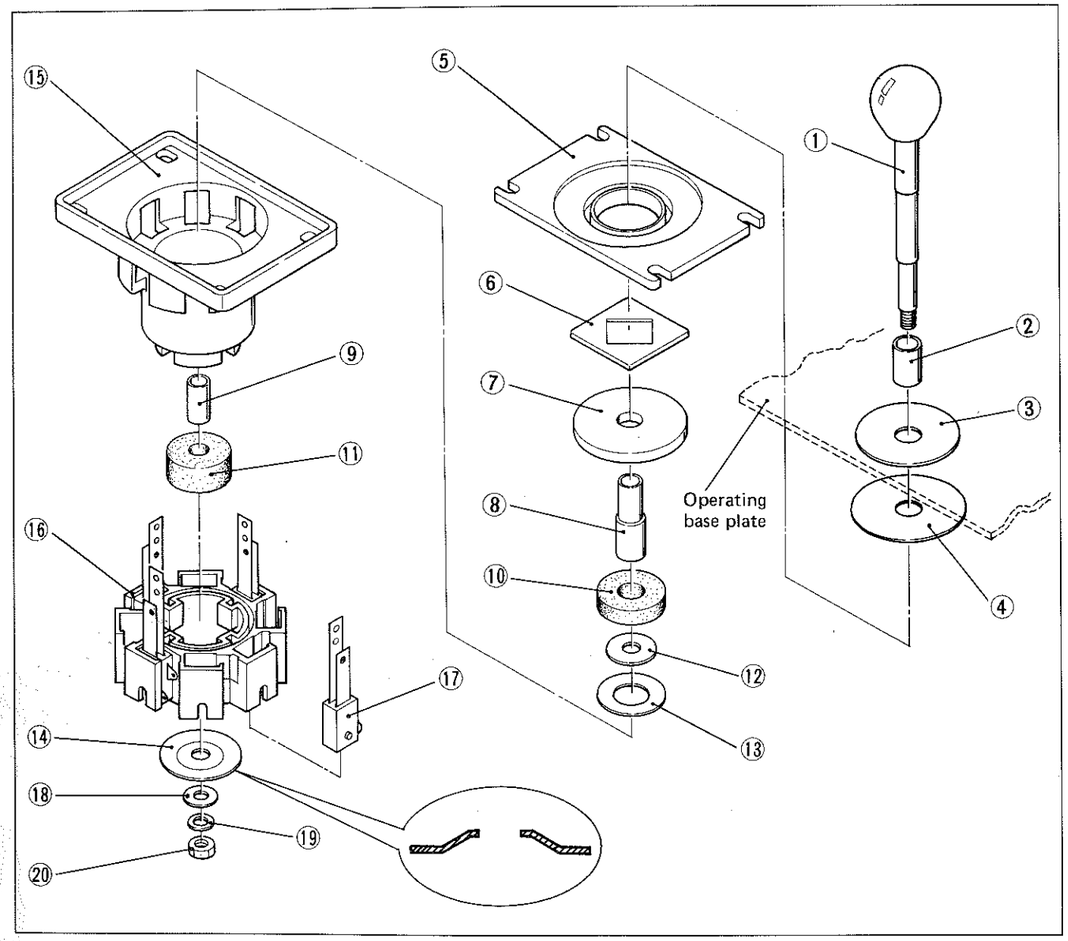


Fig. 12 Control Lever Assembly

CONTROL LEVER ASSEMBLY PARTS LIST

Ref. No.	Name	Ref. No.	Name
1	Knob shaft (25 Dim.)	11	Rubber cushion (B)
2	Collar	12	Shaft guide plate (A)
3	Top guide (plastic) plate	13	Shaft guide plate (B)
4	Seal plate	14	Shaft guide plate (C)
5	Cover	15	Housing
6	Selector plate (4 directions)	16	Switch holder
7	Actuator plate (plastic)	17	Blade switch (x 4)
8	Shaft sleeve (A)	18	Flat washer
9	Shaft sleeve (B)	19	Spring washer
10	Rubber cushion (A)	20	Lock nut

d) Push-button switch assembly and parts list

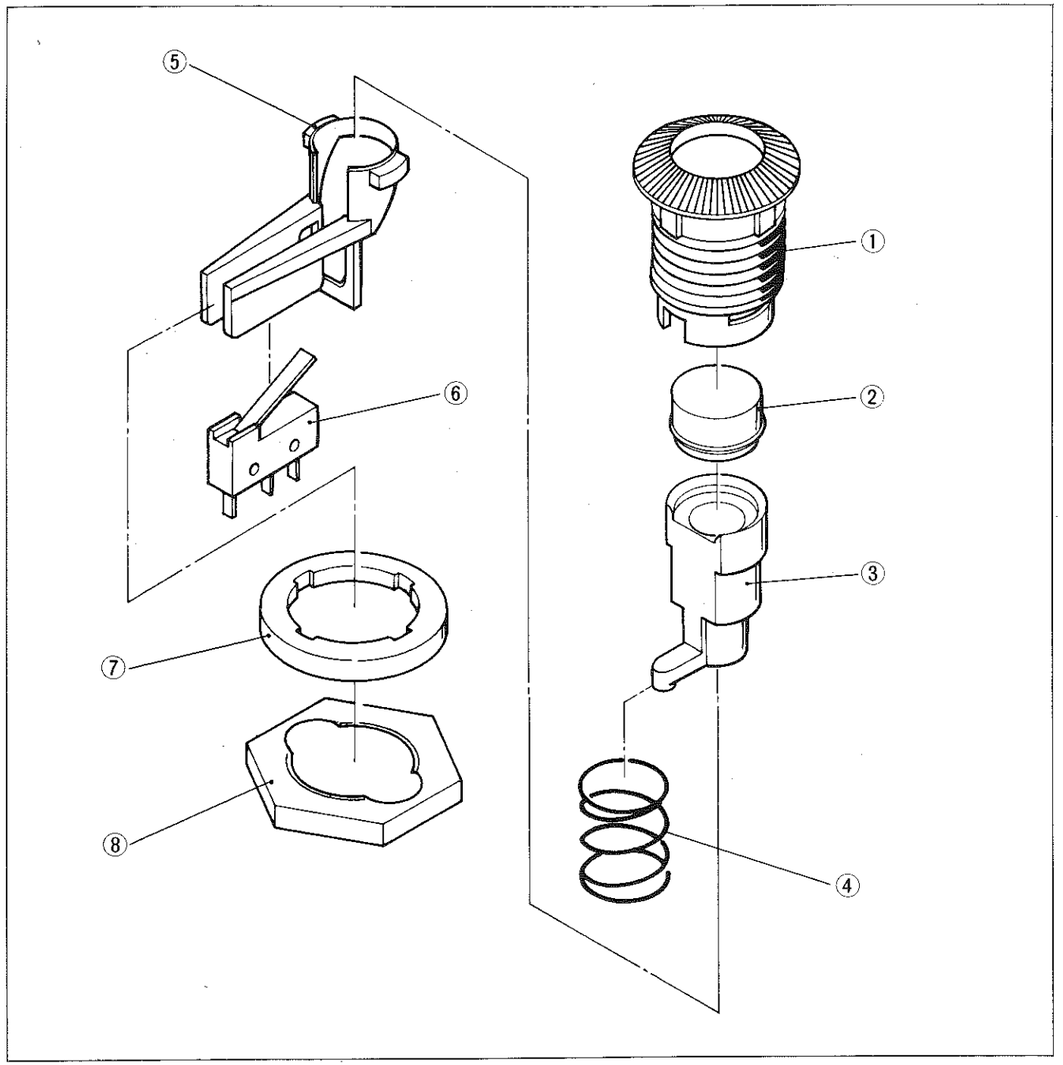


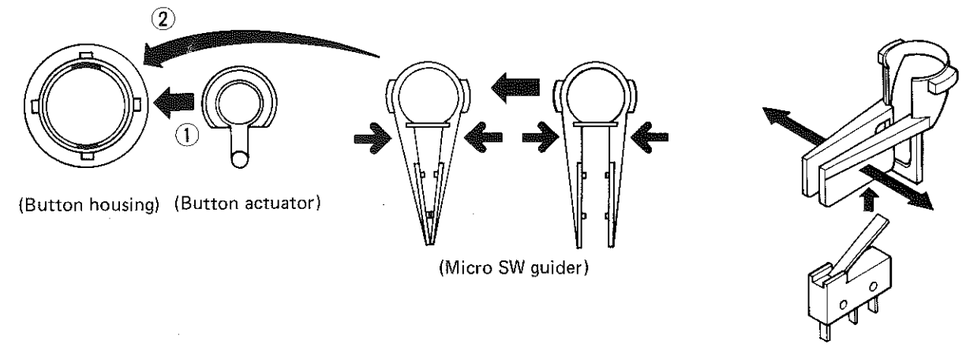
Fig. 13 Push-button Switch Assembly

PUSH-BUTTON SW ASSEMBLY PARTS LIST

Ref. No.	Name	Ref. No.	Name
1	Button housing	5	Micro switch guider
2	Button cap	6	Micro switch
3	Button actuator	7	Liner ring
4	Coil spring	8	Pearl nut

e) How to assembly control lever and push-button SW, and mount them on the operating base plate

PUSH-BUTTON SWITCH



When assembling the section shown in the drawing, be careful about the direction in which each part is assembled. In particular, when assembling the micro SW guider, depress it with your fingers from both sides at arrow-marked points, so that right- and left-hand clicks of the micro SW guider fit correctly into the side hole of the button housing.

When mounting the micro SW, widen the arm section in the arrow-marked direction, then, insert the SW between the arms. Then, leave your fingers.

Fig. 14 How to Assemble Push-button SW

HOW TO MOUNT ON THE OPERATING BASE PLATE

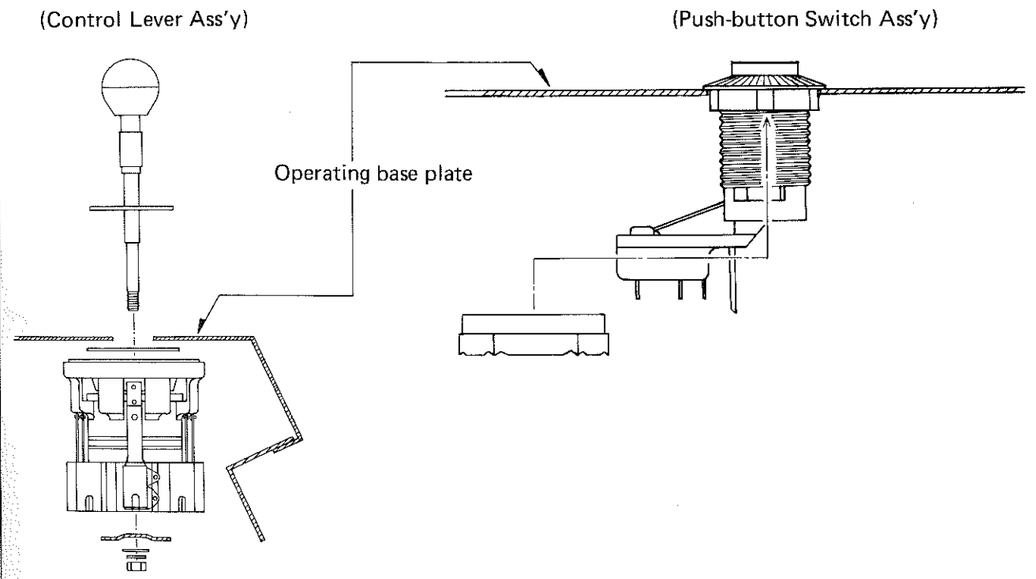


Fig. 15 How to Mount Control Lever and Push-button SW on the Operating Base Plate

E. COMPONENT PARTS RELATED TO SWITCHES IN THE CABINET

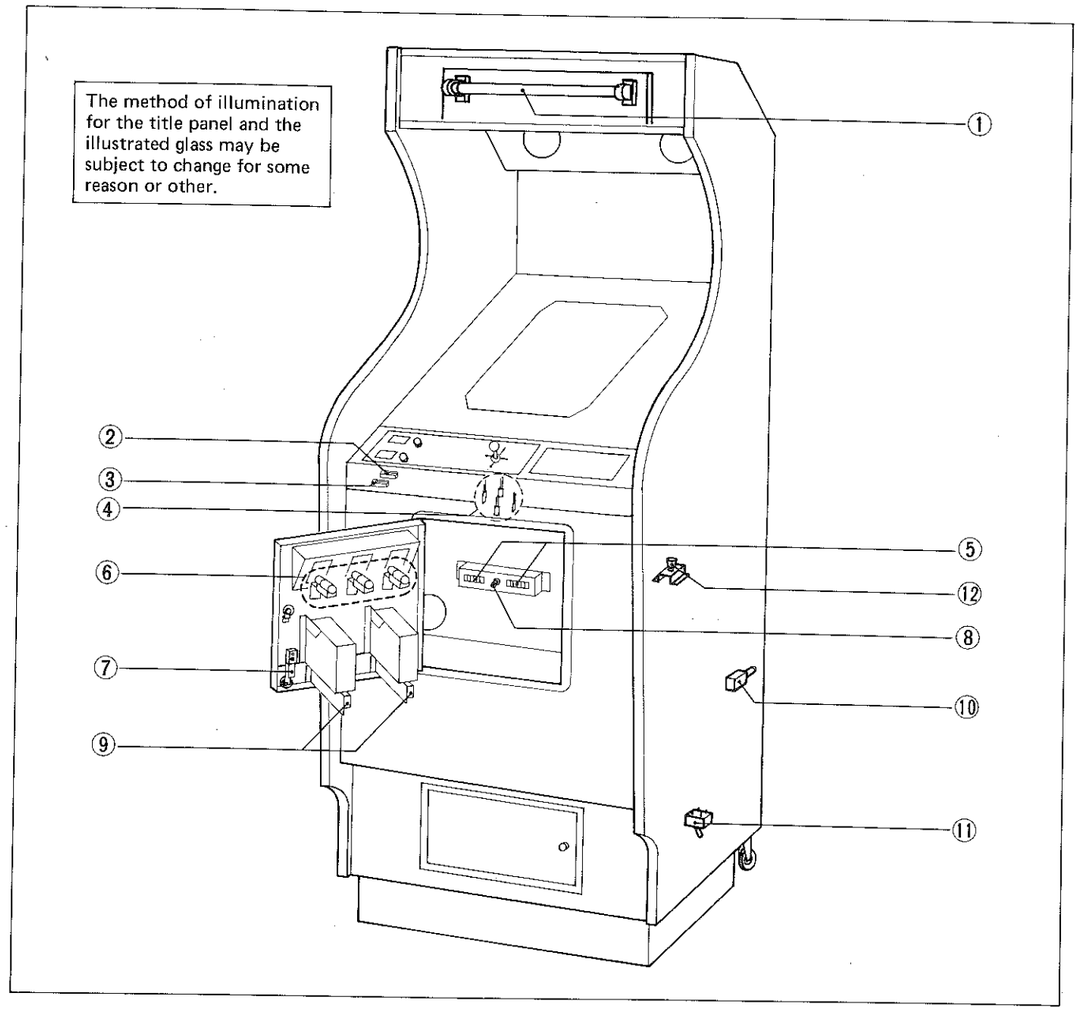


Fig. 16 Component Parts Related to Switches in the Cabinet

INSIDE CABINET PARTS LIST

Ref. No.	Name	Ref. No.	Name
1	Fluorescent lamp (for title panel)	7	Blade SW (for tilt SW)
2	1 player select button micro SW	8	Micro SW (for resetting)
3	2 players select button micro SW	9	Micro SW (for rejector x 2)
4	Control lever 4-directional use blade SW (x4)	10	Door SW
5	Coin counter (x 2)	11	Toggle SW (for main SW)
6	Miniature lamp (for money display x 3)	12	Micor SW (for demagnetizations)

TABLE TYPE



[20" Table Type are also available.]

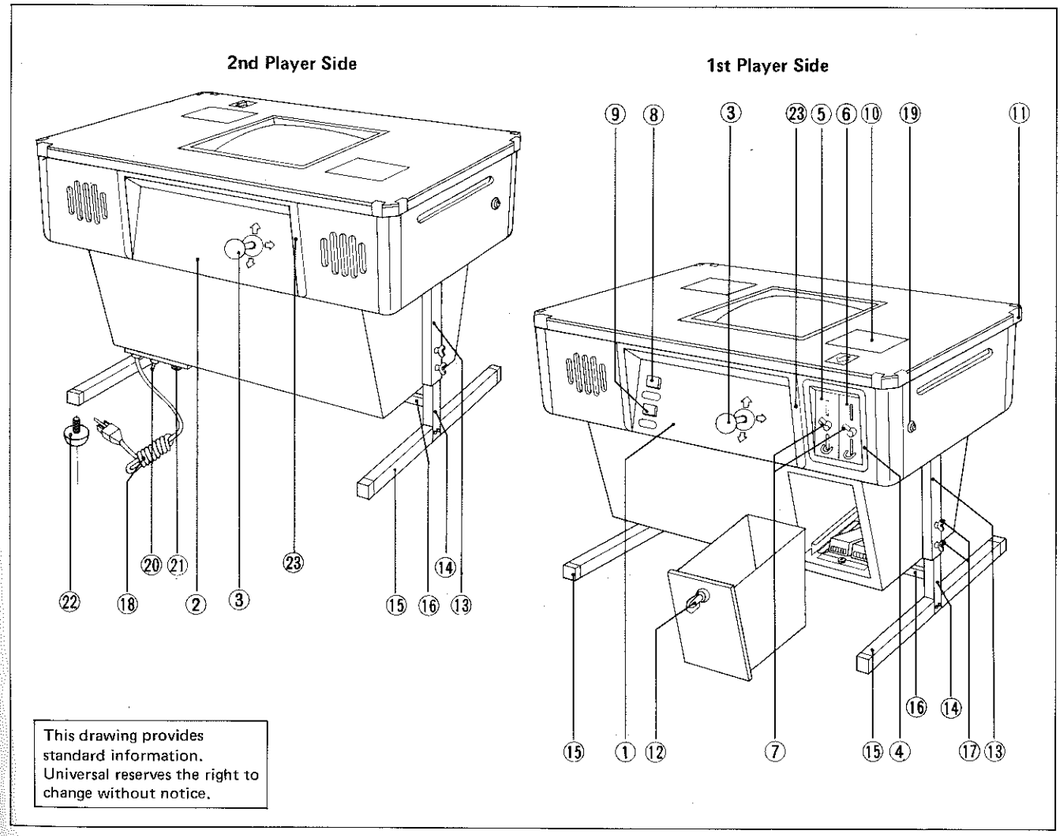
SPECIFICATIONS

▽ 95-2158
572(D) x 842(W) x 613 – 733(H)mm
AC100V/115V/230V 50/60Hz
140W (14") 180W (20")

This documentation provides standard information.
Universal reserves the right to change without notice.

VII. TABLE TYPE PARTS CATALOG

A. COMPONENT PARTS RELATED TO CABINET (OUTSIDE)



This drawing provides standard information. Universal reserves the right to change without notice.

Fig. 17 Component Parts Related to Cabinet (Outside)

OUTSIDE CABINET PARTS LIST

Ref. No.	Name	Ref. No.	Name
1	Operating indication illustrated panel A (for 1st player side)	11	Glass fittings (x 4)
2	Operating indication illustrated panel B (for 2nd player side)	12	Cash box key
3	Control lever ass'y	13	Leg (x 2)
4	Coin slot frame	14	Foot adjuster (x 2)
5	Selector plate (1)	15	Foot
6	Selector plate (2) [not used in case of 1-way]	16	Foot support
7	Cancel lever	17	Butterfly bolt
8	Push button ass'y (for 1 player)	18	Power cord (AC)
9	Push button ass'y (for 2 players)	19	Table key (x 2)
10	Sticker for game rules (x 2)	20	Toggle SW (Main)
		21	Fuse (5A for power input)
		22	Leg adjuster
		23	Operating panel frame

B. COMPONENT PARTS RELATED TO CABINET (INSIDE)

a) Inside cabinet

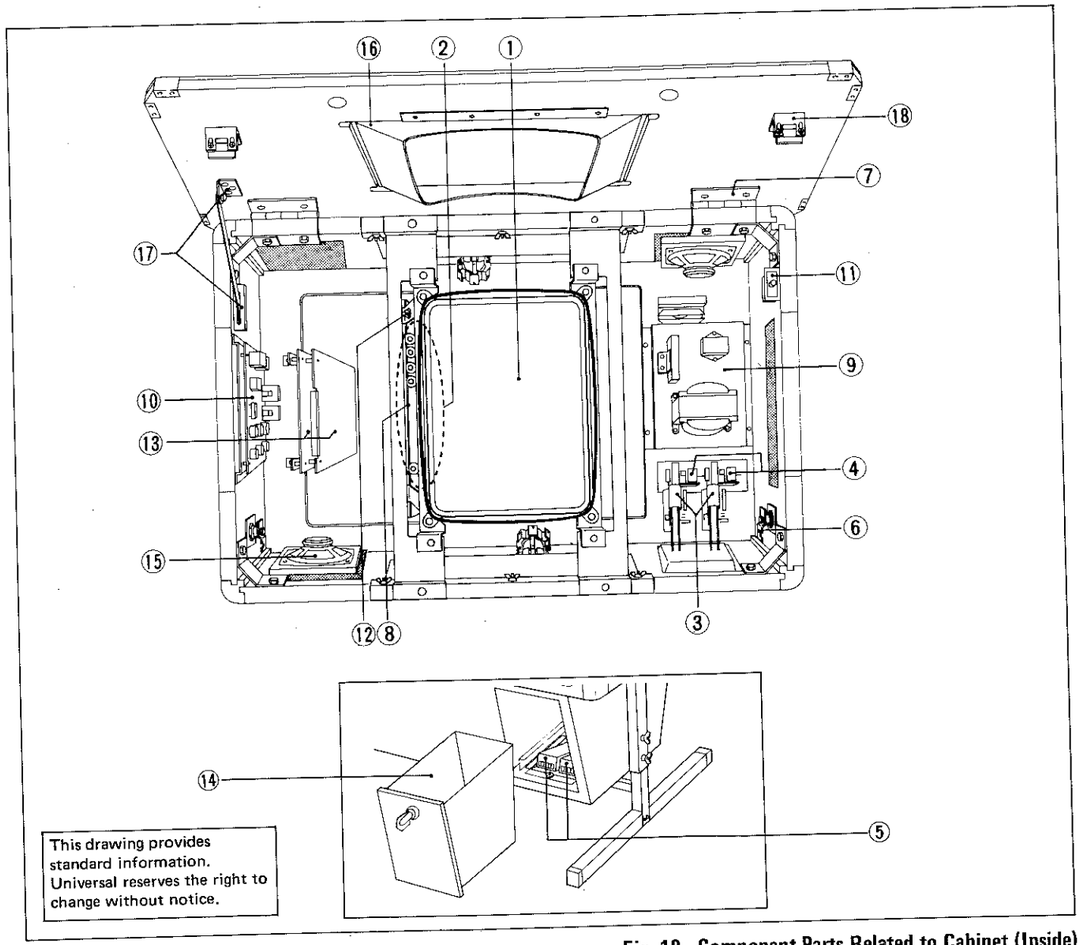


Fig. 18 Component Parts Related to Cabinet (Inside)

INSIDE CABINET PARTS LIST

Ref. No.	Name	Ref. No.	Name
1	CRT (14" color)	10	Power source circuit board
2	TV monitor control	11	Door switch
3	Rejector ass'y	12	Demagnetization switch
4	Coin micro switch	13	Main circuit board (A, B)
5	Coin counter (x2)	14	Cash box
6	Table lock hook	15	Speaker (x 2)
7	Butterfly plate ass'y	16	TV monitor mask
8	TV monitor circuit board	17	Hood guide ass'y
9	Power source unit	18	Table lock bracket

b) Fuses

Fuses in the Power Source Circuit Board

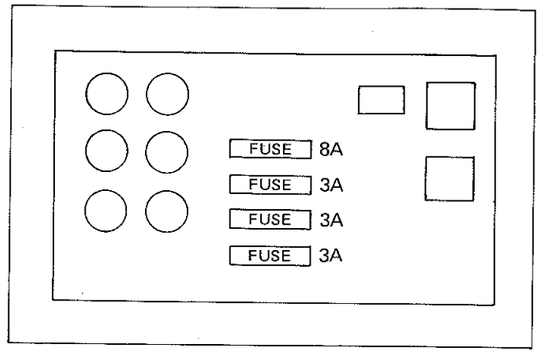


Fig. 19 Fuses in the Power Source Circuit Board

Fuse on the cabinet bottom

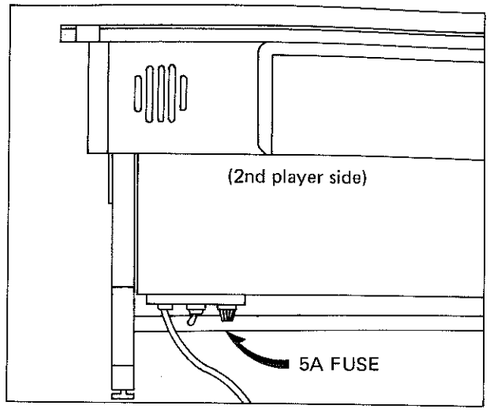


Fig. 20 Fuse on the Cabinet Bottom

c) Operating panel assembly

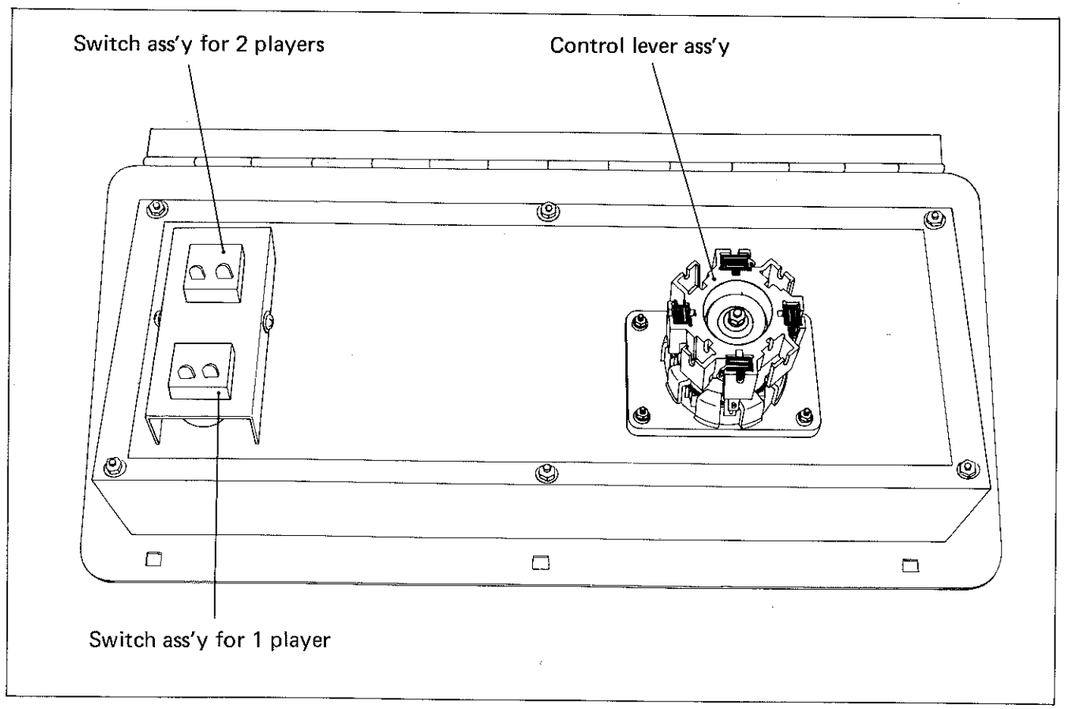


Fig. 21 Operating Panel Assembly

VIII. CIRCUIT BOARD

A. CIRCUIT BOARD IC LOCATION AND PARTS LIST

a) Main circuit board IC location and parts list

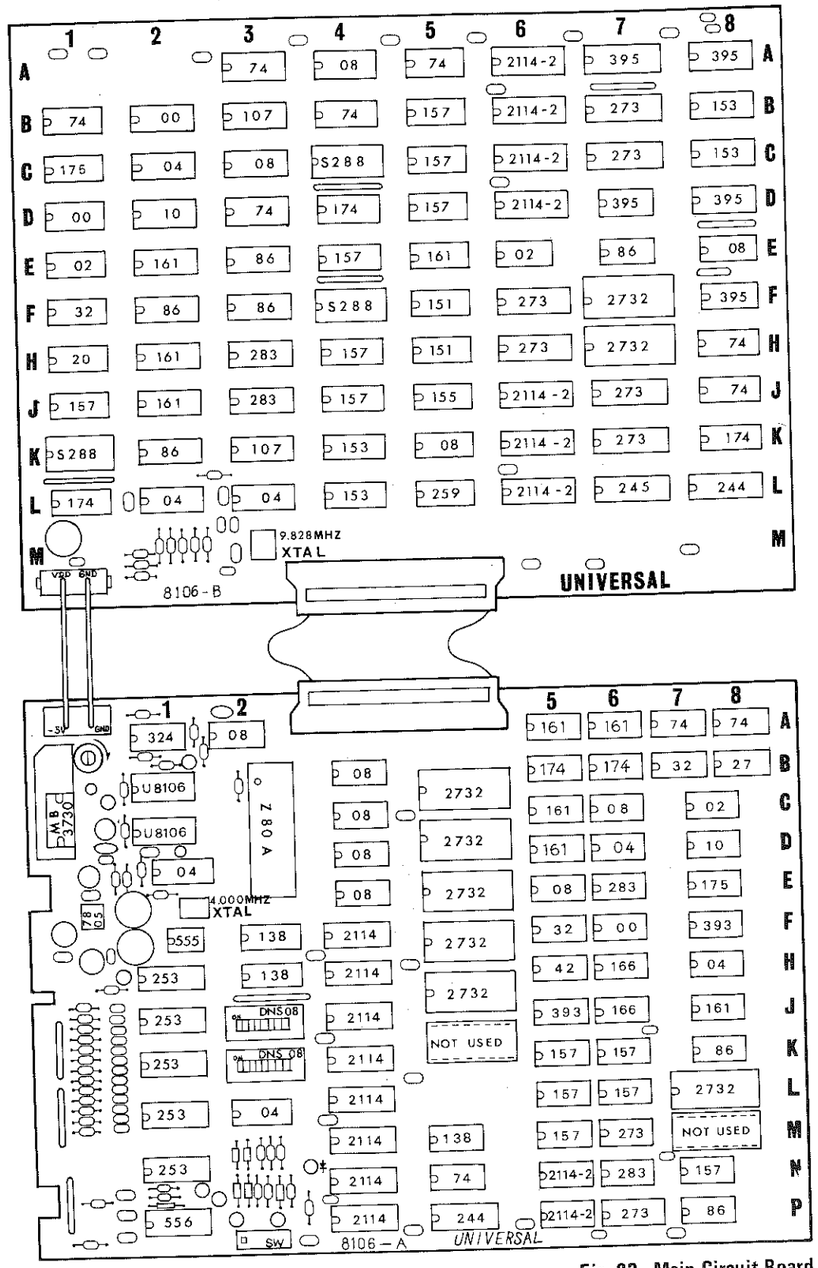


Fig. 22 Main Circuit Board

This drawing provides standard information. Universal reserves the right to change without notice.

[1] Integrated Circuit

Item No.	Q'ty	Description
74LS 00	3	TTL
74LS 02	3	"
74LS 04	5	"
74 04	1	"
74 S 04	1	"
74LS 08	11	"
74LS 10	2	"
74LS 20	1	"
74LS 27	1	"
74LS 32	3	"
74LS 42	1	"
74LS 74	9	"
74LS 86	7	"
74LS107	2	"
74LS138	3	"
74LS151	2	"
74LS153	4	"
74LS155	1	"
74LS157	13	"
74LS161	9	"
74LS166	2	"
74LS174	6	"
74LS175	2	"
74LS244	2	"
74LS245	1	"
74LS253	5	"
74LS259	1	"
74LS273	8	"
74LS283	4	"
74 S288 (TBP18S030)	3	256 bits Bipolar RAM
74LS393	2	TTL
74LS395	5	"
NE555	1	Timer
NE556	1	"
Z80A	1	Nch MOS CPU (4MHz)
2732	3	Nch MOS 32K bits EP ROM (Access 450nsec)
2732-35	5	Nch MOS 32K bits EP ROM (Access 350nsec)
2114	8	Nch MOS 4K bits Static RAM (Access 450nsec)
2114-2	9	Nch MOS 4K bits Static RAM (Access 250nsec)
U8106	2	Special function
LM324	1	Quad Operational Amplifiers
MB3730	1	Power Amplifiers
MA7805	1	Regulator

[2] Other Semiconductor Devices

Item No.	Q'ty	Description
TLR103	1	LED
10D1	6	Diode

[3] Capacitors

Rating	Q'ty	Description
100P1/12V	2	Ceramic Capacitor
0.001μ/ "	1	"
0.01μ/ "	2	"
0.1μ / "	54	"
0.1μ/25V	3	"
1μ/25V	1	Chemical Capacitor
47μ/25	2	"
4.7μ/25V	2	"
10μ/ "	2	"
22μ/ "	1	"
220μ/ "	4	"
470μ/ "	2	"

[4] Registors

Rating	Q'ty	Description
20Ω¼W	1	Carbon Solid
2KΩ¼W	2	"
100Ω¼W	2	"
4.7Ω "	2	"
47Ω "	1	"
220Ω "	18	"
330Ω "	2	"
470Ω "	6	"
510Ω "	4	"
1KΩ "	1	"
4.7KΩ "	1	"
5.1KΩ "	1	"
10KΩ "	2	"
47KΩ "	1	"
51KΩ "	1	"
100KΩ "	3	"
MS1028AM	8	1KΩ Registor Array
MS1024AM	1	"
MS1038AM	2	10KΩ Registor Array
RV16YP or RGP102	1	1KΩ (B) Variable Registor

[5] Misc

Name	Q'ty	Description
Dip SW	2	8 Elements Switch Array
X tal	2	9.828MHz
		4.000MHz
E61-00A	1	Micro SW

WIRING DIAGRAM (CONNECTOR)

UPRIGHT TYPE

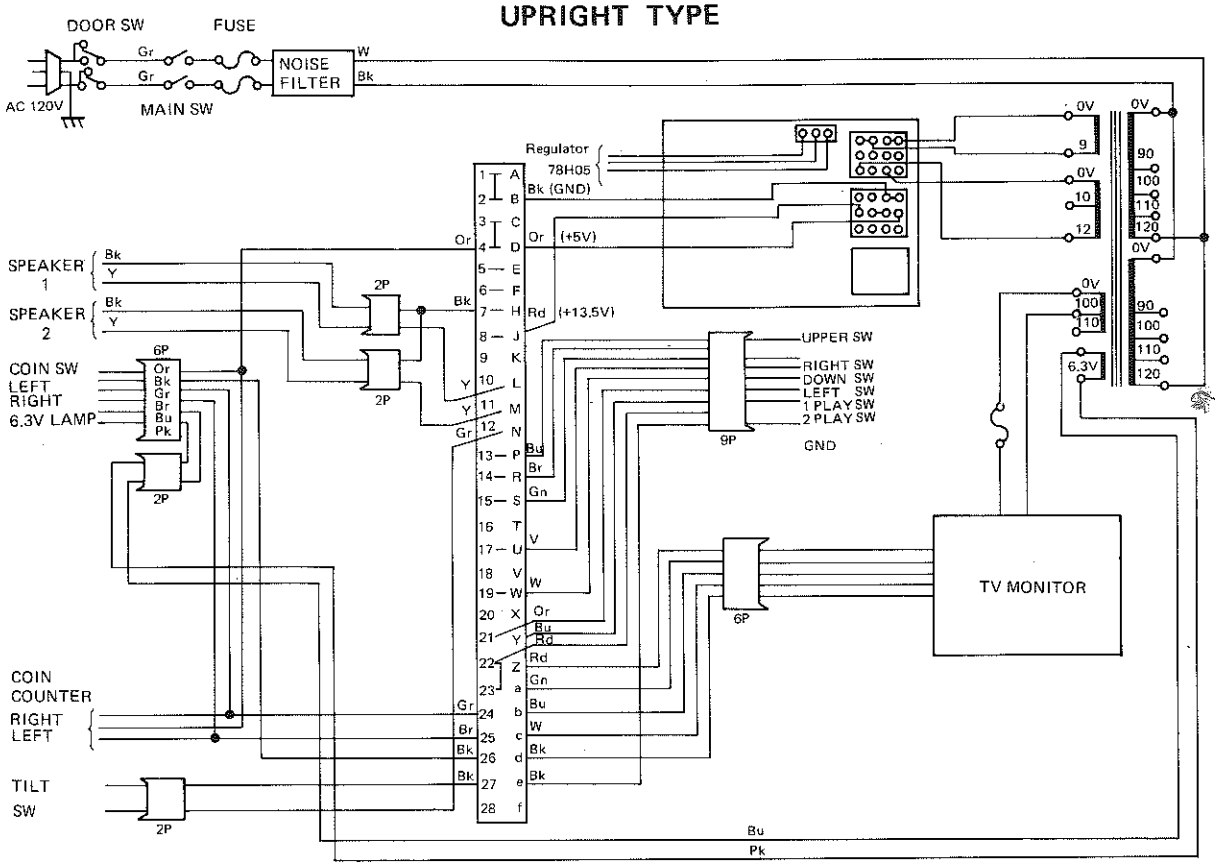
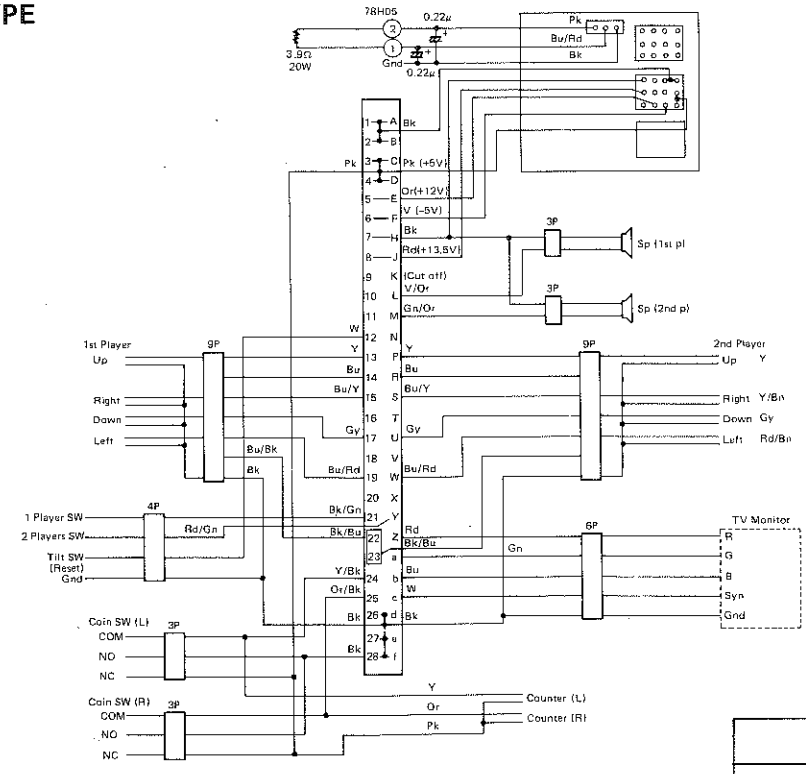


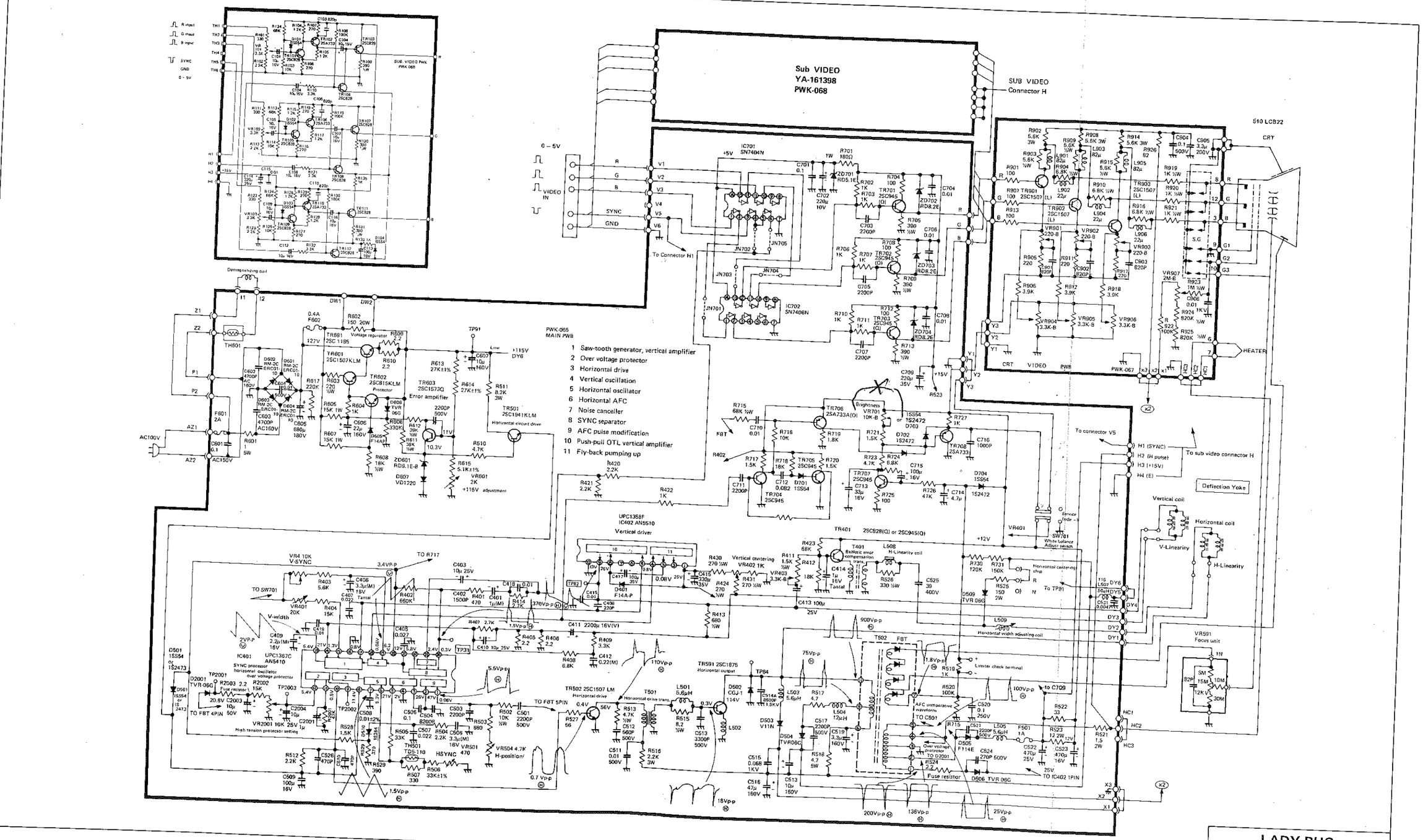
TABLE TYPE



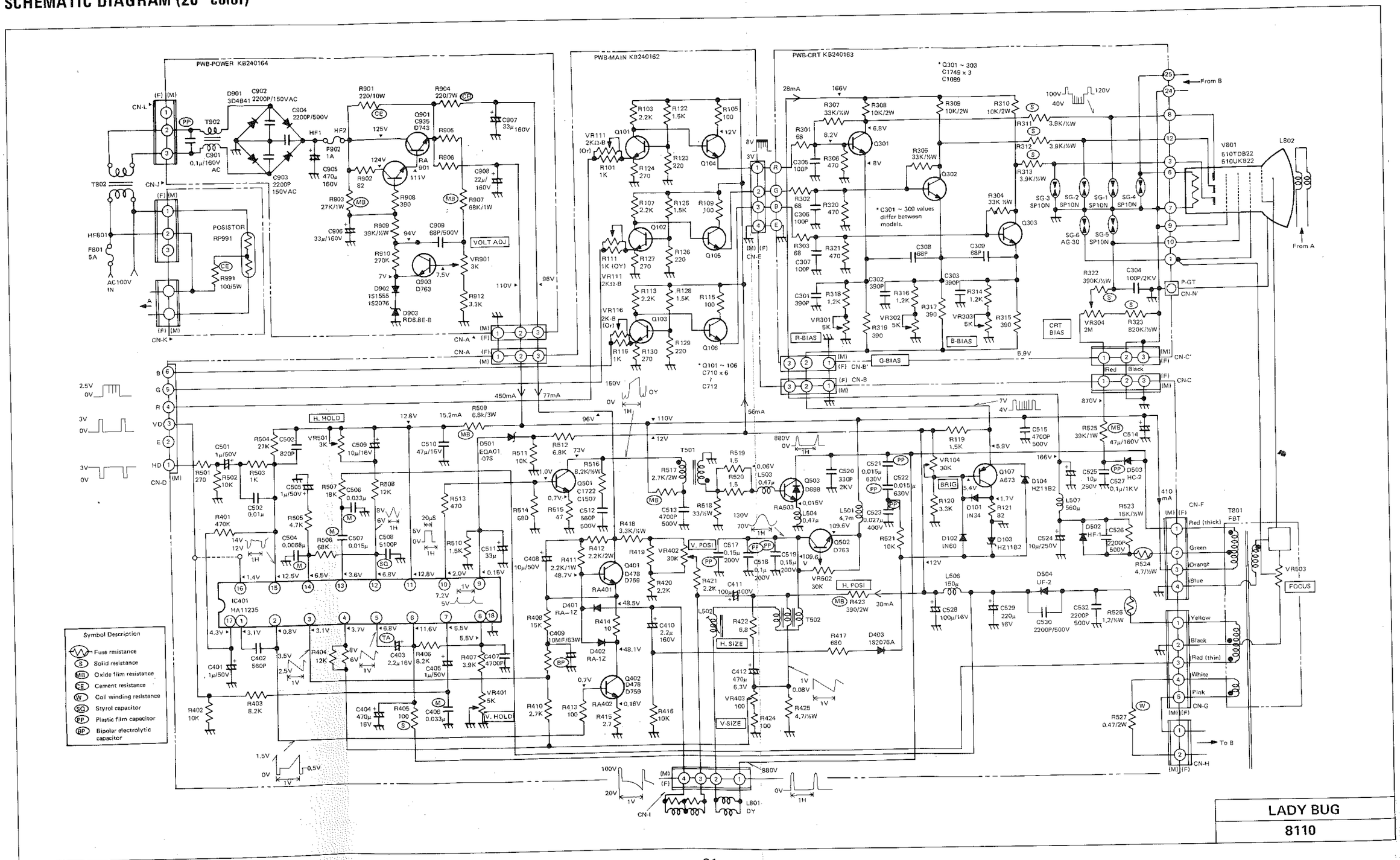
LADY BUG

8110

SCHEMATIC DIAGRAM (20" color)

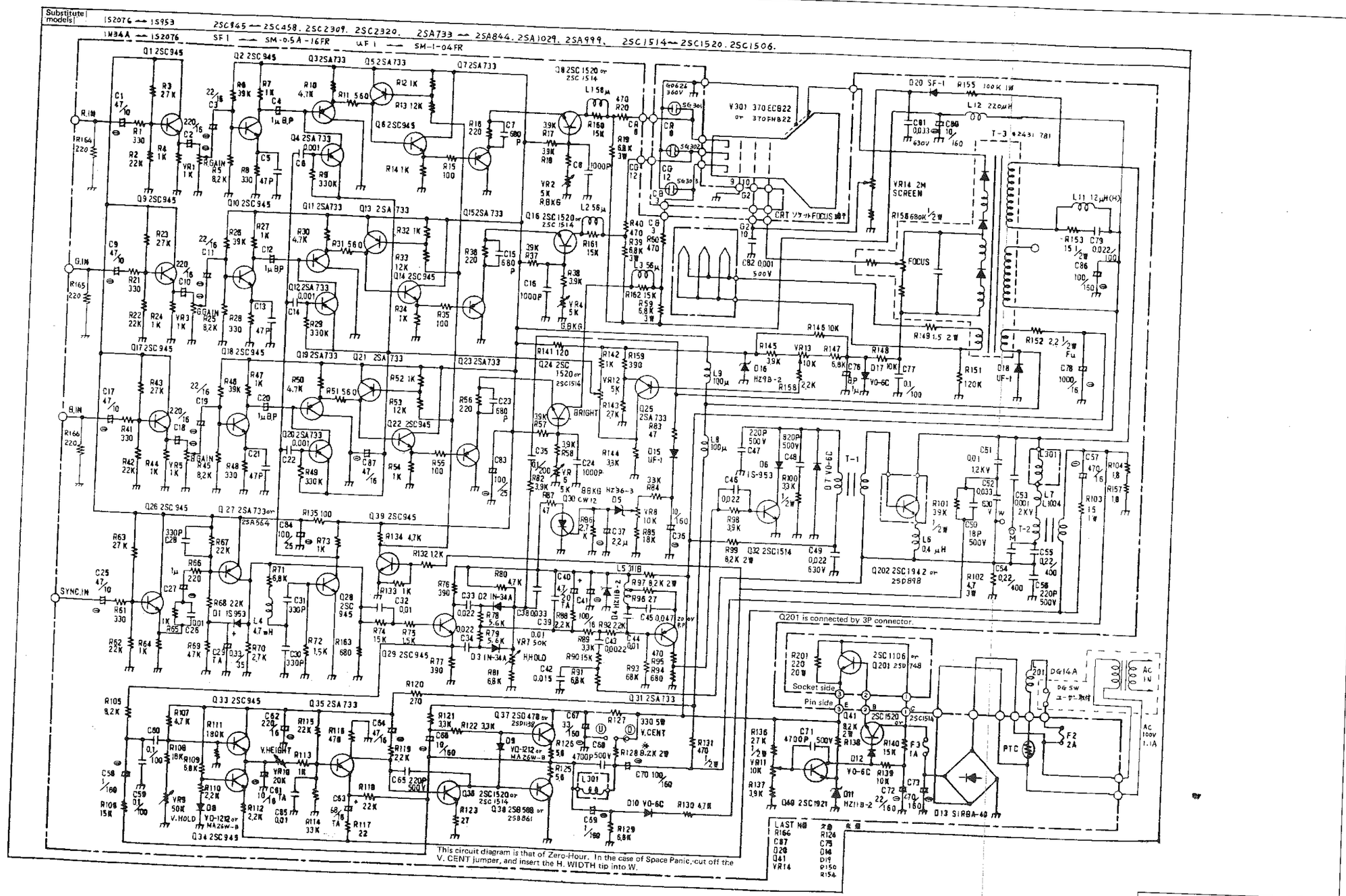


SCHEMATIC DIAGRAM (20" color)



LADY BUG
8110

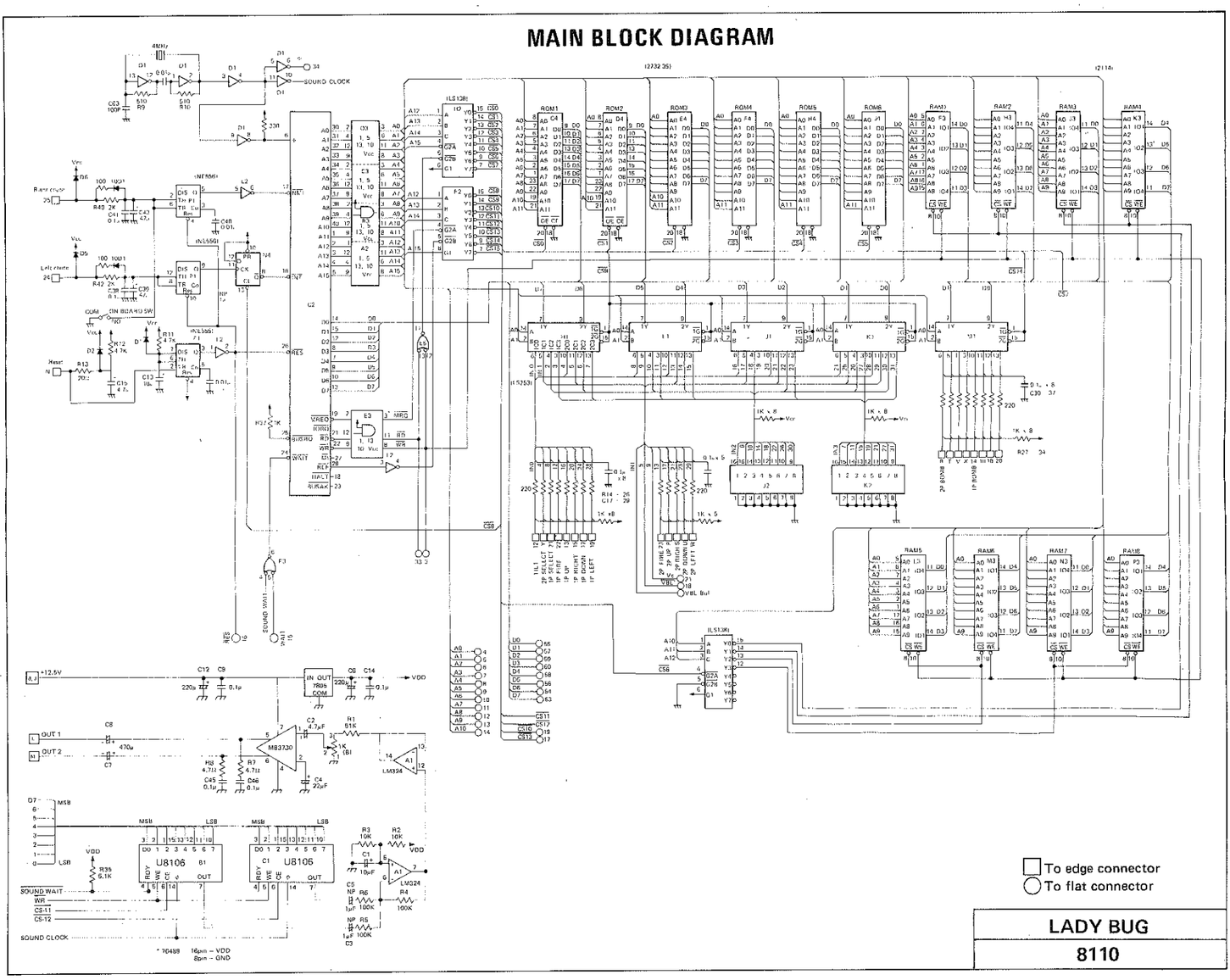
SCHEMATIC DIAGRAM (14" color)



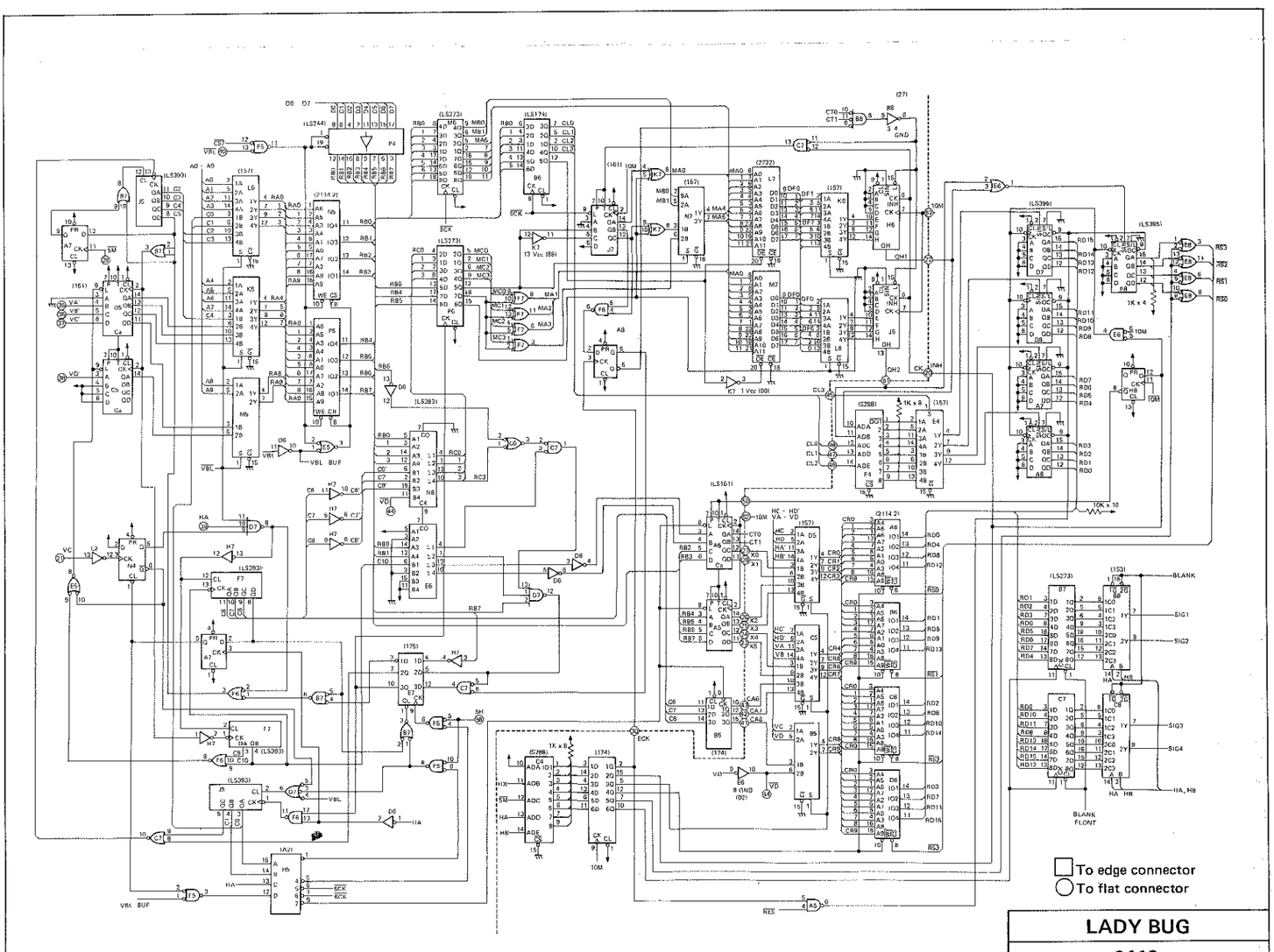
This circuit diagram is that of Zero-Hour. In the case of Space Panic, cut off the V. CENT jumper, and insert the H. WIDTH tip into W.

LADY BUG
8110

MAIN BLOCK DIAGRAM

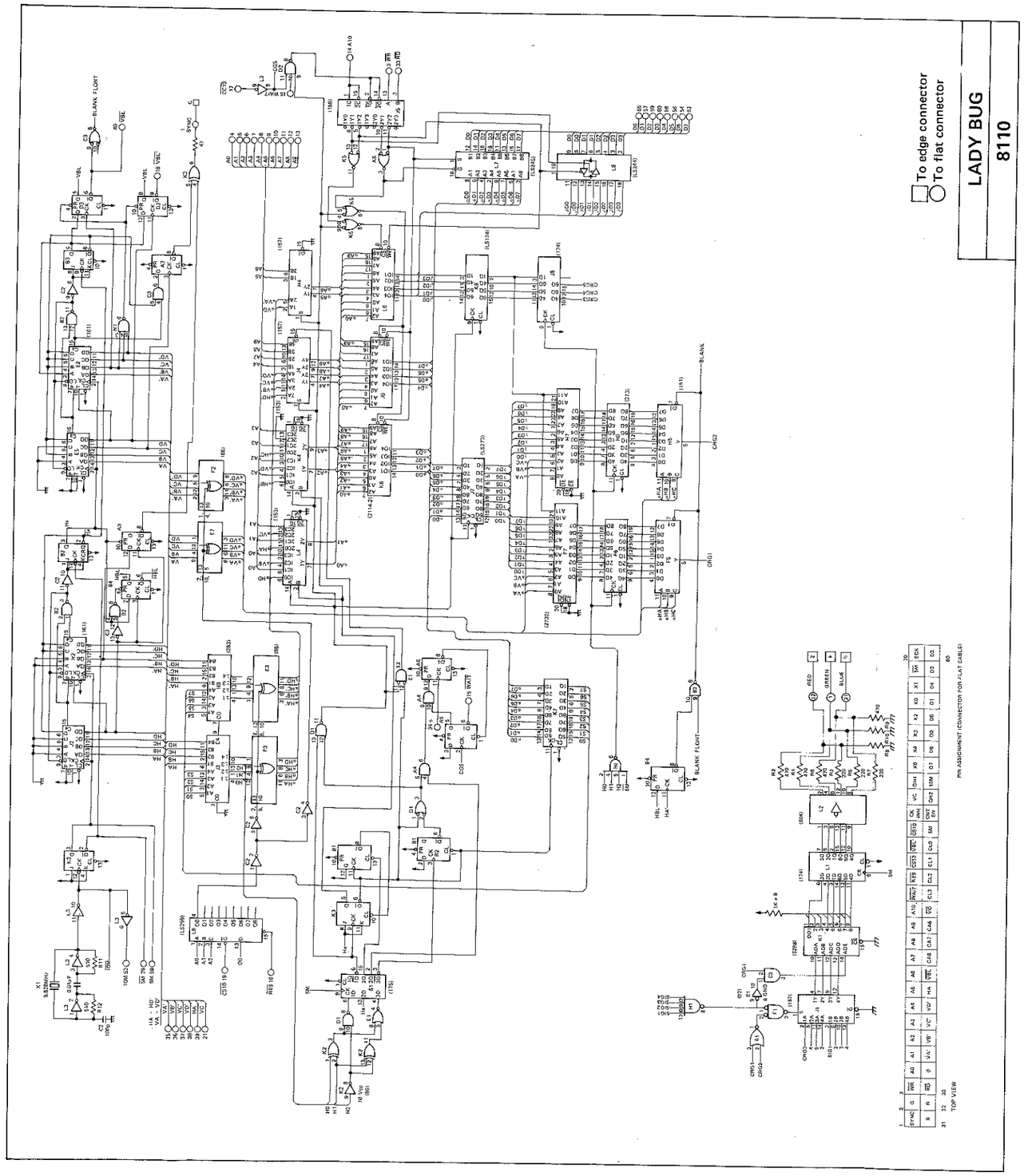


LADY BUG
8110

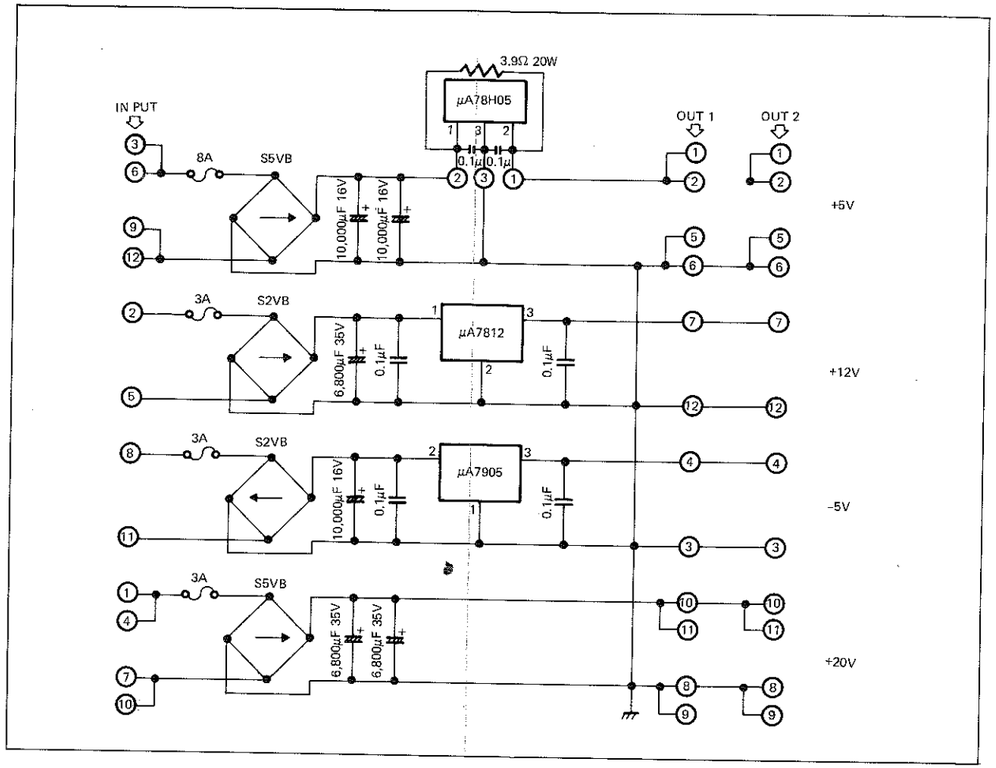


LADY BUG
8110

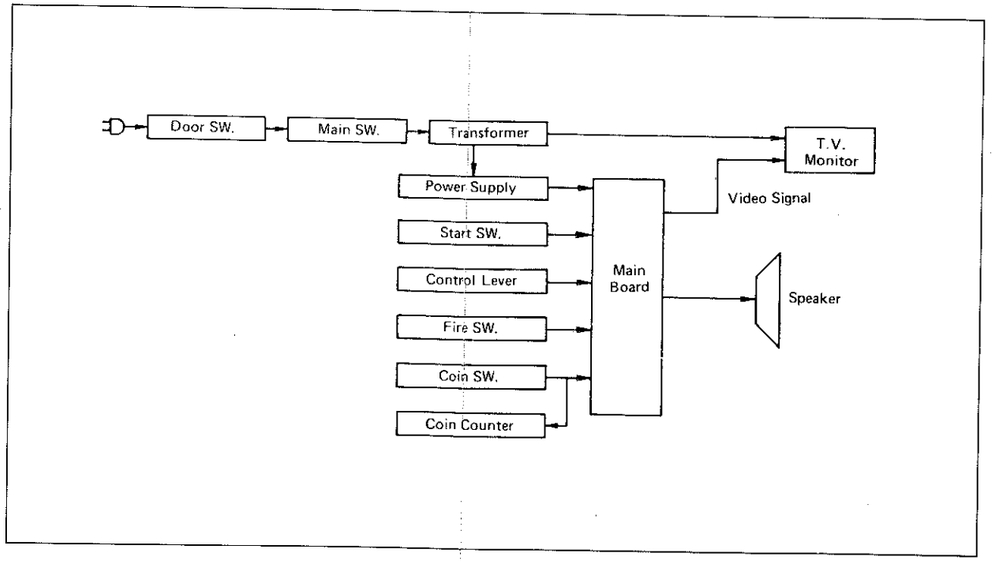
UNIVERSAL



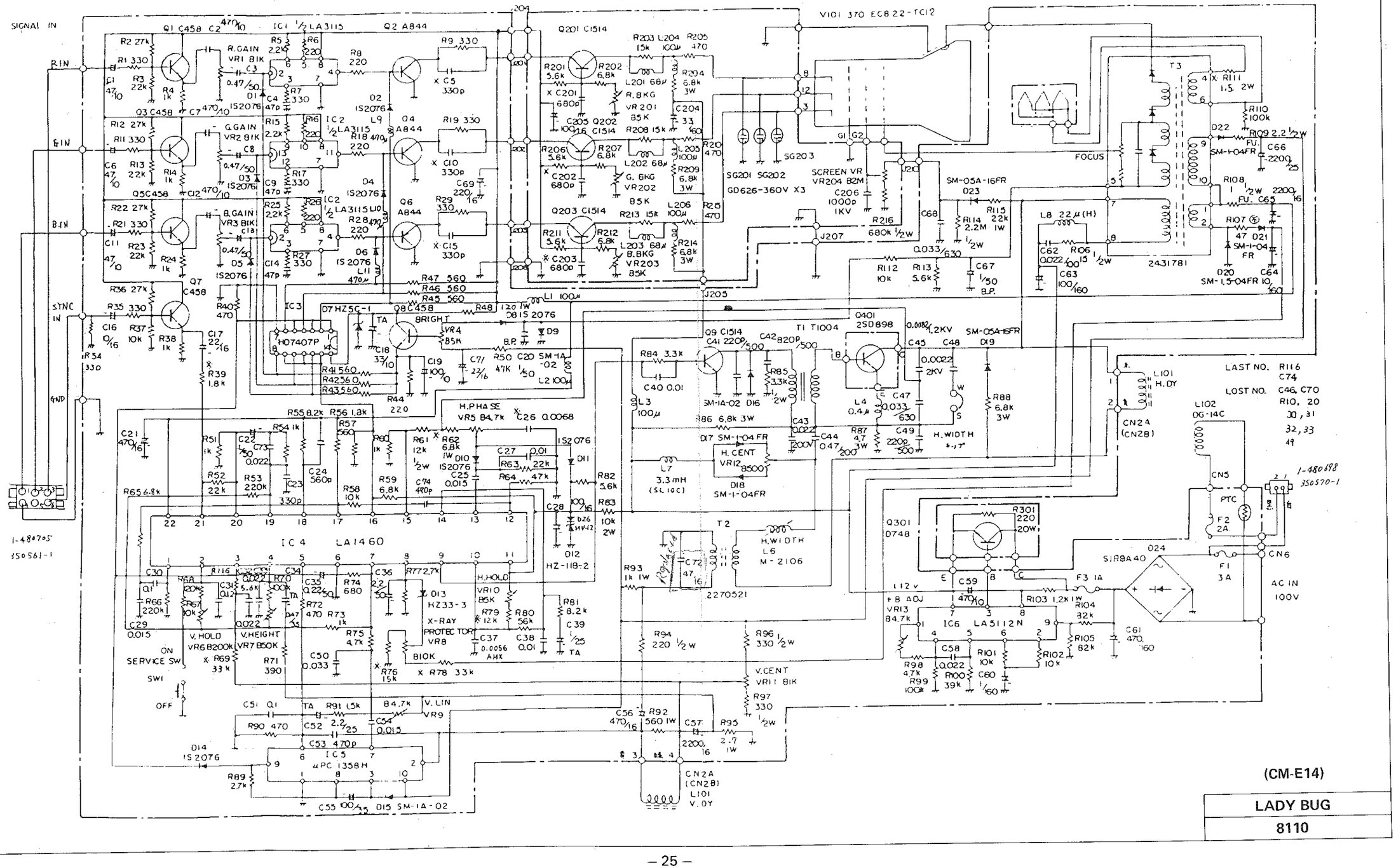
POWER SOURCE BLOCK DIAGRAM

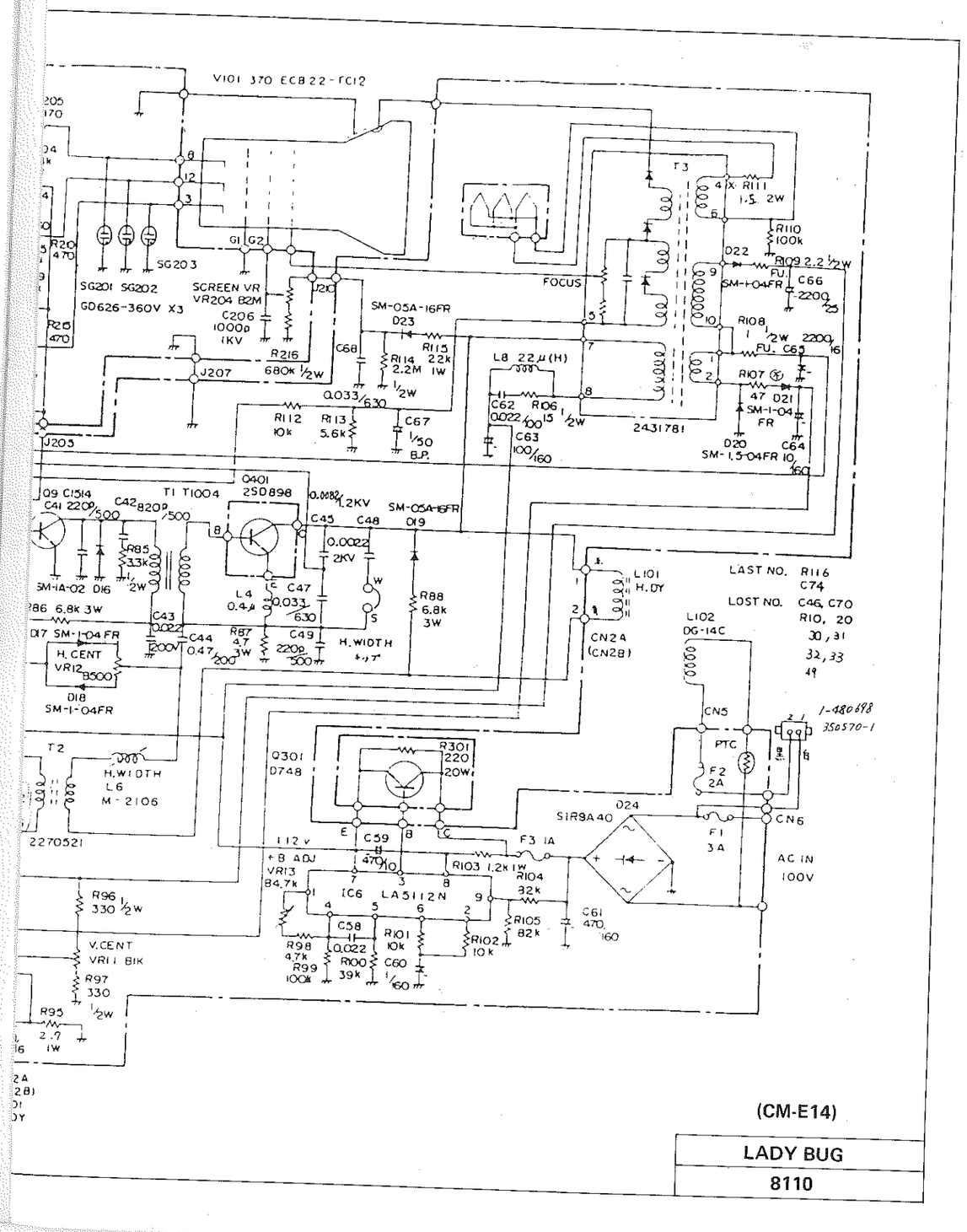


BLOCK DIAGRAM



SCHEMATIC DIAGRAM (14" color)





UNIVERSAL

CAUTION (to prevent X-ray exposure from the CRT)

Two variable resistors, the +B ADJ VR (VR11 8-1K ohm) and the X-ray protector VR (VR8 B-10K ohms) are found within the monitor.

They are rigidly adhered for both safety and to prevent X-ray exposure from the CRT.

(IMPORTANT:Please do not remove or change them!)

When a component has to be replaced due to damage to the VR(s) please follow the following procedure when exchanging it.

(IMPORTANT: The monitor must be synchronized with the signal when adjusting.)

(1) To set for +B ADJ VR (VR11)
+B line voltage is to be set at 107V DC ±1V when adjusting VR11.

(2) To set for X-ray protector VR8

(2)-1. Place VR in a counter clockwise position to stop it before turning the monitor power switch to "ON" and decrease AC power supply voltage by approximately 10%.

(2)-2. To be shorted for resistor of 220 ohms 20W R201 which is installed on heat sink plate.

(2)-3. Power supply should be placed in the "ON" position to keep the surface of the CRT slightly fluorescent – i.e. bright.

(2)-4. +B line voltage is to be set at 123V DC±1V while increasing AC power supply voltage.

(2)-5. When the VR8 to be fixed is in the horizontal oscillation position it may be stopped by slowly rotating it clock wise.

(2)-6. Once the power supply switch is turned "OFF" and AC power has decreased approximately 10% it can be turned "ON" again. Horizontal oscillation may then be confirmed by stopping +B line voltage at 123V DC±1V while slowly increasing AC voltage.

(3) The VR must again be rigidly adhered thus making any further adjustments impossible.

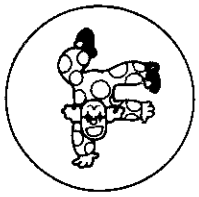
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Please note that the specifications may be subject to change along with quality improvement.

MEMO

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